

Bayesian Neural Networks

General Principles

To model complex, non-linear relationships between variables, we can utilize various approaches, including [splines](#), [polynomials](#), [Gaussian processes](#), and neural networks. Here, we focus on the *Bayesian Neural Network (BNN)*. Think of a neural network as a highly flexible function composed of interconnected layers of neurons. Each connection possesses a *weight*, and each neuron has a *bias*. These *weights* and *biases* act as a vast set of adjustable knobs. In a standard neural network, the goal is to find the single best setting for these knobs to map inputs to outputs. Conversely, a BNN learns **distributions** over its *weights* and *biases* rather than single optimal values. This allows the model to capture not only the patterns in the data but also its own uncertainty regarding those patterns.

To construct a *BNN*, we must define:

- 1) **A Network Architecture:** This specifies the structural design—the number of layers, the number of neurons per layer, and the activation functions (e.g., ReLU, tanh) that introduce non-linearity. This defines the arrangement of our “knobs.”
- 2) **Priors for Arrays of Weights and Biases:** In simple models like linear regression, we define a prior for each individual parameter (e.g., the slope β). However, neural networks often contain thousands or millions of parameters, making it impractical to define a unique prior for each one. Instead, we define priors that act as templates for entire **arrays of parameters**. For instance, we might specify that all weights in a given layer are drawn from the same $\text{Normal}(0, 1)$ distribution. This allows us to efficiently quantify our beliefs about the entire set of network parameters.
- 3) **An Output Distribution (Likelihood):** This defines the probability of the observed data given the network’s predictions. For continuous variables (regression), this is typically a *Normal distribution* with a variance term σ , which quantifies the noise of the data around the model’s predictions.

Considerations

Caution

- **Uncertainty:** Like all Bayesian models, BNNs explicitly account for model parameter uncertainty . Here, the parameters are the network’s **weights (W)** and **biases (b)**. We quantify uncertainty through their posterior distribution . Consequently, we must declare prior distributions for all *weights* and *biases*, as well as for the output variance σ .
- **Interpretation:** Unlike in a linear regression where the coefficient β has a direct interpretation (e.g., the change in Y given a unit increase in X), the individual *weights* and *biases* in a *BNN* are not directly interpretable. A single weight’s influence is entangled with thousands of others through non-linear functions. Therefore, BNNs should be viewed as powerful **predictive tools** rather than explanatory ones. They excel at learning complex topologies and quantifying predictive uncertainty, but if the goal is to isolate the effect of a specific variable, a simpler model is often more appropriate.
- **Prior distributions:** are built following these considerations:
 - As the data is typically scaled (see introduction), we can use a standard Normal distribution (mean 0, standard deviation 1) as a weakly-informative prior for all weights and biases. This acts as a form of regularization.
 - Since the output variance σ must be positive, we can use a positively-defined distribution, such as the Exponential or Half-Normal.
- BNNs can be used for both *regression* and *classification*. The final layer’s activation and the chosen likelihood distribution depend on the task. For binary classification, a *sigmoid* activation is paired with a Bernoulli likelihood, which requires a link function (logit) to connect the linear output of the network to the probability space $[0, 1]$. For regression, the identity activation is often used with a Gaussian likelihood.
- **Computational Scalability:** Strictly applying Bayes’ theorem to update the posterior distribution becomes computationally intractable for very large neural networks. Calculating the model evidence (the denominator in Bayes’ rule) requires integrating over a parameter space that may contain millions of dimensions. Furthermore, the posterior landscape in deep networks is often highly non-convex and multimodal. Consequently, sampling algorithms (such as MCMC) may suffer from poor mixing or become “stuck” in local modes, failing to explore the full posterior. In such high-dimensional scenarios, practitioners often must rely on approximate methods, such as Variational Inference or Stochastic Gradient MCMC , to ensure convergence.

Example

Below is an example code snippet demonstrating a *Bayesian Neural Network* for regression using the Bayesian Inference (BI) package. Simulated data consist of two continuous variables (Y and X), and the goal is to predict X from Y using a non-linear model.

Python

```
from BI import bi
import json
import jax.numpy as jnp
import matplotlib.pyplot as plt

# Setup device-----
m = bi(platform='cpu')

# Import Data & Data Manipulation -----

with open('BNN.json', 'r', encoding='utf-8') as file:
    # Load the JSON data into a Python dictionary
    data = json.load(file)
# X is already scaled
X = jnp.array(data['X']) # Note X shape = (N,2) where first column is the intercept and second column is the feature
Y = jnp.array(data['Y']) # Note Y shape = (N,1) where N is the number of observations

m.data_on_model = dict(X = X, Y = Y)
# Define model -----
def model(X, Y, D_H=5, D_Y=1):
    N, D_X = X.shape

    # First hidden layer: Transforms input to  $N \times D_H$  (hidden units)
    w1 = m.bnn.layer_linear(
        X,
        dist=m.dist.normal(
            0, 1, name='w1', shape=(D_X, D_H)
        ),
        activation='tanh'
    )

    # sample final layer of weights and neural network output
    # Final layer (z3) computes linear combination of second hidden layer
```

```

w2 = m.bnn.layer_linear(
    X=w1,
    dist=m.dist.normal(0, 1, name='w2',shape=(D_H,D_Y))
)

sigma = m.dist.exponential(1, name='sigma')

m.dist.normal(w2, sigma, obs=Y,name='Y')

# Run mcmc -----
m.fit(model, progress_bar=False) # Approximate posterior distributions for weights, biases

# Predictions from the model -----
pred = m.sample(samples = 500)['Y']
pred = pred[..., 0]
mean_prediction = jnp.mean(pred, axis=0)
percentiles = jnp.percentile(pred, jnp.array([5.0, 95.0]), axis=0)
# make plots
fig, ax = plt.subplots(figsize=(8, 6), constrained_layout=True)
# plot training data
ax.plot(X[:, 1], Y[:, 0], "kx")
# plot 90% confidence level of predictions
ax.fill_between(
    X[:, 1], percentiles[0, :], percentiles[1, :], color="lightblue"
)
# plot mean prediction
ax.plot(X[:, 1], mean_prediction, "blue", ls="solid", lw=2.0)
ax.set(xlabel="X", ylabel="Y", title="Mean predictions with 90% CI")

```

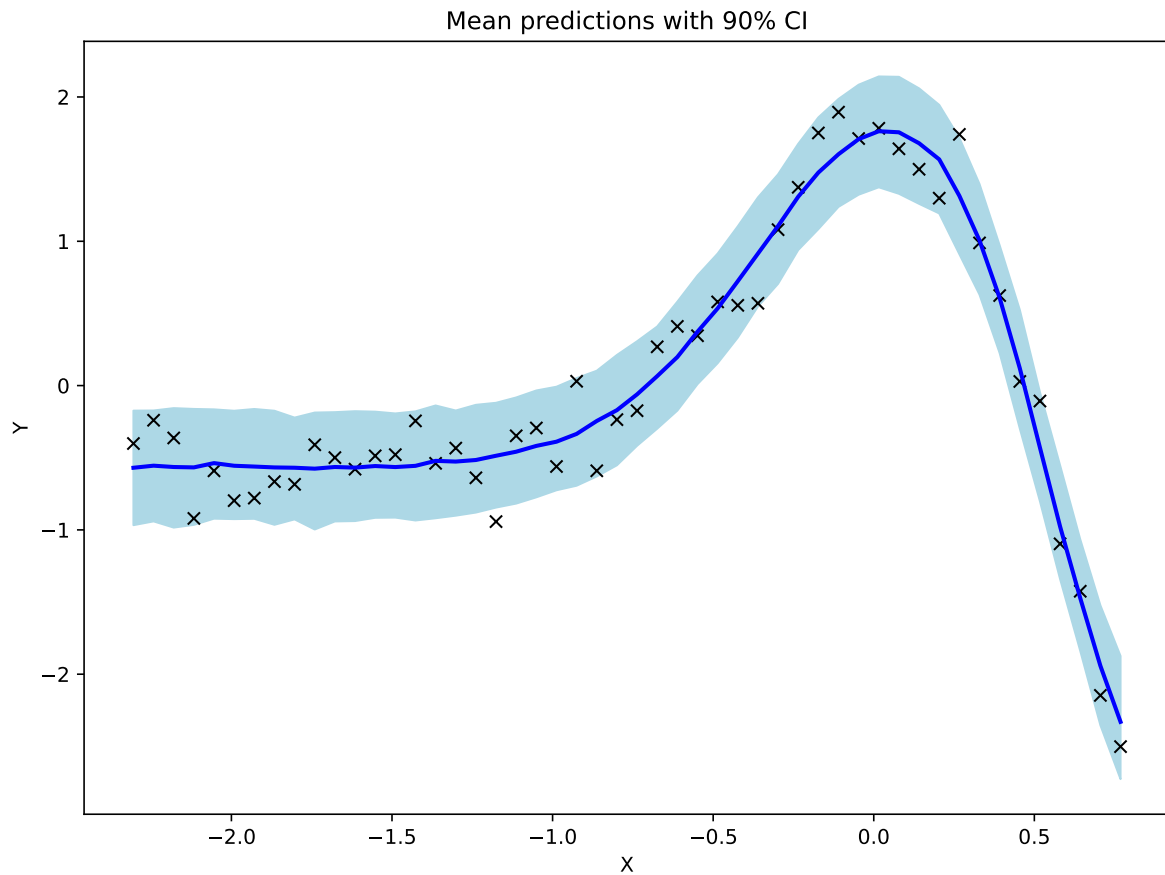
jax.local_device_count 16

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```

[Text(0.5, 0, 'X'),
Text(0, 0.5, 'Y'),
Text(0.5, 1.0, 'Mean predictions with 90% CI')]

```



R

```
library(BayesianInference)
m=importBI(platform='cpu')

# Load csv file
# Import data -----
path = normalizePath(paste(system.file(package = "BayesianInference"),"/data/BNN.json", sep = "/"))
data <- fromJSON(path)
m$data_on_model = list()
m$data_on_model$X = jnp$array(data$X)
m$data_on_model$Y = jnp$array(data$Y)

# Define model -----
model <- function(X, Y, D_X = 2, D_H=5L, D_Y=1L){
```

```

w1 <- m$bnn$layer_linear(X, dist=bi.dist.normal(0, 1, name='w1',shape=c(D_X,D_H)), activa

w2 <- m$bnn$layer_linear(
  w1,
  dist=bi.dist.normal(0, 1, name='w2',shape=c(D_H,D_Y)),
  activation='tanh'
)

# Prior for the output standard deviation
s = bi.dist.exponential(1, name = 's')

# Likelihood
bi.dist.normal(w2, s, obs = Y)
}

# Run mcmc -----
m$fit(model) # Approximate posterior distributions

```

Mathematical Details

In the Bayesian formulation, we place priors on all weights and biases and define a likelihood for the output. For a regression task with a one-hidden-layer BNN with J neurons per hidden layer and a 5-vector of predictors and a 2-vector of outcomes we can run the model as below.

For a given individual i assuming you have data on their height H_i , weight W_i , age A_i , and strength S_i , the the covariate vector (including the intercept of 1) can be written as:

$$X_i = [1, H_i, W_i, A_i, S_i]$$

We can map this into a hidden layer as follows. Let Θ_1 be a $J \times 5$ matrix of weights and b_1 be a 5-vector of biases, then the first hidden layer can be written as:

$$\Omega_{i,1} = f(X_i\Theta_1 + b_1)$$

Next, we can had an arbitrary number of hidden layers. In this case, we will include only a single additional layer, $\Omega_{i,2}$. In this case, Θ_2 is an $J \times J$ matrix of weights and b_2 is a J -vector of biases. This second hidden layer can be written as:

$$\Omega_{i,2} = f(\Omega_{i,1}\Theta_2 + b_2)$$

Finally we can map to the output layer. In this case, Θ_3 is a $J \times 2$ matrix of weights and b_3 is a 2-vector of biases. This leads to a final output layer mapping to the dimensions of the observed outcome (in this a 2-vector):

$$\Omega_{i,3} = f(\Omega_{i,2}\Theta_3 + b_3)$$

Finally, we can model the observed outcomes using $\Omega_{i,3}$ as the mean vector of a normal distribution, with a standard deviation of σ . This is the likelihood function for the output layer:

$$\begin{aligned}\mu &= \Omega_{i,3} \\ Y_{i,1} &\sim \text{Normal}(\mu_1, \sigma_1) \\ Y_{i,2} &\sim \text{Normal}(\mu_2, \sigma_2)\end{aligned}$$

With priors for σ :

$$\begin{aligned}\sigma_1 &\sim \text{Exponential}(1) \\ \sigma_2 &\sim \text{Exponential}(1)\end{aligned}$$

Notes

Note

- The primary difference between a *Frequentist* and *Bayesian* neural network lies in how parameters are treated. In the frequentist approach, weights and biases are point estimates found by minimizing a loss function (e.g., via gradient descent). Techniques like Dropout or L2 regularization are often used to prevent overfitting, which can be interpreted as approximations to a Bayesian treatment. In contrast, the *Bayesian* formulation does not seek a single best set of weights. Instead, it uses methods like MCMC or Variational Inference to approximate the entire posterior distribution for every *weight* and *bias*. This provides a principled and direct way to quantify model uncertainty.
- While present an example of non-linear regression, the Bayesian Neural Network can be used for linear regressions as well (keeping in mind that interpretation of the weights are impossible).

Reference(s)

Phan, Pradhan, and Jankowiak (2019)

Phan, Du, Neeraj Pradhan, and Martin Jankowiak. 2019. “Composable Effects for Flexible and Accelerated Probabilistic Programming in NumPyro.” *arXiv Preprint arXiv:1912.11554*.